

Cerio

Project Documentation

October 7, 2026

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1 Overview

The subject of the project is the development of a two-dimensional platform game in the style of Mario or Celeste (reference subject to change) and, as its main part, the design of algorithms and models for automatic level completion by a bot.

2 Game rules

The main goal of the game is to reach the exit. The game consists of procedurally generated levels. Death sends the player back to the start of the level. Each level contains:

- a level entrance and exit,
- an area in which the player can move,
- stationary obstacles capable of killing the player.

Some levels also contain interactive elements, e.g. buttons. Completing a level activates a checkpoint, so it does not have to be repeated.

2.1 Game goals

Main goal: reaching the level exit.

Intermediate goals: avoiding obstacles, activating interactive elements, managing stamina, reaching checkpoints.

Bonus goals: collecting bonus points (coins).

Failure condition: contact with an obstacle, which restarts the level.

2.2 Game mechanics

- wall climbing,
- traversing dynamic levels,
- avoiding obstacles,
- stamina mechanics,
- interaction with the environment (buttons, levers),
- elements that kill the character,
- bonus points (coins) and other collectibles.

Available movement actions: dash, move left, move right, jump, crouch, climb.

3 Bot

3.1 Operation

The bot runs inside the game, without the graphics layer. Instead of an image, it reads game metadata: the level layout, obstacles, interactive elements and the character state (position, velocity, stamina).

Based on this data, it selects successive actions and moves towards the exit. Decisions are made either by an algorithm searching the space of possible moves or by a trained model.

The bot completes previously unseen procedural levels.

3.2 Planner

The planner is a state-space search algorithm (A^*) that, before moving, computes a sequence of actions leading to the exit. For the current game state it simulates the effects of the available actions, discards moves that end in death or lead to already visited states, and expands the states closest to the exit first. Deterministic game physics is a necessary condition.

The planner serves three purposes:

1. it is the v1 bot (algorithmic approach),
2. it verifies that procedurally generated levels can be completed,
3. it provides demonstrations for training the v2 model (PyTorch).

4 Requirements

4.1 Performance requirements

Minimum hardware configuration: AMD Ryzen 5 7600X (6 cores), NVIDIA GeForce RTX 3070, 48 GB RAM. Resolution 854×480 px, maintained for 80% of the gameplay, assuming the game is “playable”.

4.2 Evaluation criteria

- The bot must complete at least 70% of a test set consisting of 10 levels.
- The bot must complete the task in finite time, shorter than three times the human completion time.

4.3 Technology stack

Python, Pygame, PyTorch.

5 Schedule

Date	Checkpoint	Scope
08.10.2026	Project requirements	Requirements, game goals, bot assumptions
22.10.2026	Report	Game engine basics, deterministic physics, level format
05.11.2026	Milestone I	Full game mechanics, environment interface, bot v1 (planner)
19.11.2026	Report	Procedural generator, human time measurement, test set
03.12.2026	Deadline	Bot v2, evaluation, optimization, final report

Table 1: Project checkpoints

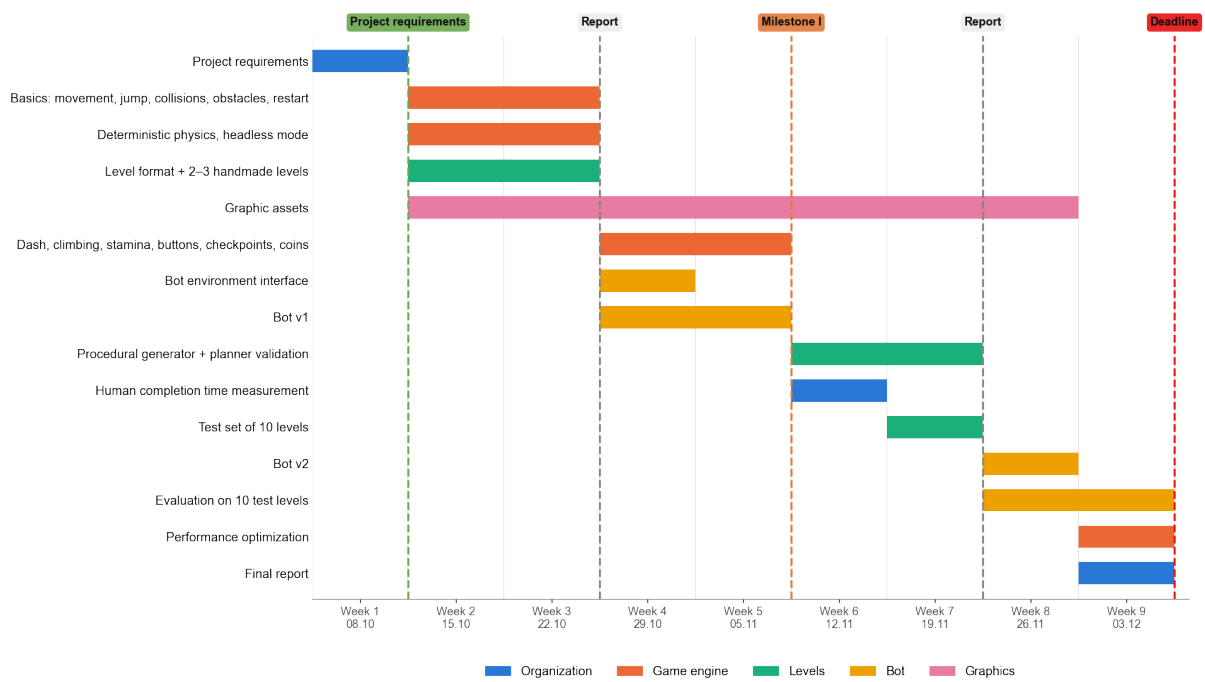


Figure 1: Gantt chart